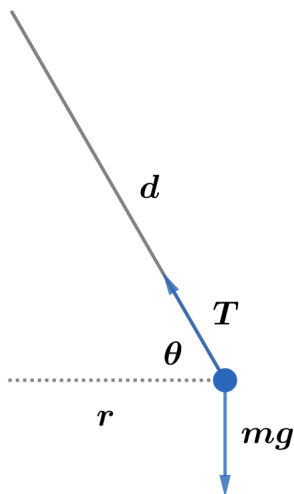


2020B F=ma Exam: Problem 3

Kevin S. Huang



The particle is at rest in the vertical direction so

$$T \sin \theta = mg$$

For circular motion,

$$T \cos \theta = m\omega^2 r$$

Using the first equation,

$$\frac{mg}{\tan \theta} = m\omega^2 r$$

Since $\tan \theta = \sqrt{d^2 - r^2}/r$,

$$\omega^2 = \frac{g}{r \tan \theta} = \frac{g}{\sqrt{d^2 - r^2}}$$

so the answer is D.